

UNIVERSITI MALAYA

EXAMINATION FOR THE DEGREE OF MASTER OF DATA SCIENCE

ACADEMIC SESSION 2024/2025 : SEMESTER 2

WQD7006/7012 : Machine Learning for Data Science/Machine Learning Application

June 2025

Time : 2 hours

INSTRUCTIONS TO CANDIDATES :

*Answer **ALL** questions. (30 marks)*

QUESTION

(Kertas soalan ini mengandungi 3 soalan dalam 4 halaman yang bercetak)
(This question paper consists of 3 questions on 4 printed pages)

Online assessment/30 marks – 2 hours

Submission through Spectrum. Specific instructions to be given on Spectrum as well.

Question 1: 10 marks

(a) A hospital plans to use a machine learning model to predict patient readmission within 30 days post-discharge, aiming to optimize care and reduce costs. Two machine learning models are available for predicting patient readmissions.

- Model A achieves 95% accuracy with low computational cost.
- Model B achieves 96% accuracy but incurs a very high computational cost.

Critically assess why the high accuracy reported by either model may still indicate overfitting or poor generalization, particularly in the medical domain. **(2 marks)**

Then, recommend and justify two alternative evaluation strategies or metrics that better reflect real-world reliability and efficiency in healthcare settings.

(3 marks)

(b) A company wants to predict if an email is spam or not spam based on the presence of the word "free". From previous data:

10 out of 15 spam emails contain the word free.
2 out of 10 non-spam emails contain the word free.
Total emails: 25 (15 spam, 10 non-spam)

Using Naive Bayes (no smoothing), calculate the probability that an email containing the word free is spam.

(3 marks)

(c) A tech company claims its AI-based email filter is 99% accurate in detecting spam. However, only 1 in 100 emails is actually spam.

If the filter flags an email as spam, is it highly likely to be spam? Briefly justify your answer using a Bayesian perspective.

(2 marks)

Question 2 – 10 marks

(a) A public health department wants to build a machine learning model to predict the number of hospital admissions for respiratory illness based on air pollution levels (PM2.5 in $\mu\text{g}/\text{m}^3$). Using historical data, a linear regression model was trained and produced the following equation:

$$\text{Predicted Admissions} = 120 + 2.8 \times (\text{PM2.5})$$

- i. What is the model predicting when the PM2.5 level is $40 \mu\text{g}/\text{m}^3$?
(1 mark)
- ii. In the context of machine learning, what do the values 120 and 2.8 represent?
(2 marks)
- iii. List two limitations of using this model for real-world prediction.
(2 marks)

(b) A data analyst at a streaming platform is using logistic regression to predict whether a user will fully watch (1) or stop early (0) when starting a new movie. Two models are developed using different feature sets (e.g., watch history, genre preference, device type, time of day, etc.).

Metric	Model A	Model B
Accuracy	0.88	0.84
Precision (Positive=1)	0.76	0.85
Recall (Positive=1)	0.70	0.65
F1-Score	0.73	0.73
AUC-ROC	0.81	0.88

- i. Interpret the strengths and weaknesses of each model based on the metrics provided
(3 marks)
- ii. Which model would you deploy in a real-world streaming platform, and why? Consider factors such as user satisfaction, recommendation quality, and platform engagement.
(2 Marks)

Question 3: 10 marks

In a school, the headmaster is attempting to analyze the student performance, based on 3 information collected:

- a) Travel time: home to school travel time (Short, Long)
- b) Health: current health status (numeric: from 1 - very bad to 5 - very good)
- c) Absence: number of school absences (numeric: from 0 to 12)

The data collected are given in Table 1.

Table 1: Dataset for student performance analysis

Travel time	Health	Absence	Student performance
long	3	4	Pass
short	2	2	Pass
short	5	2	Fail
long	5	4	Fail
short	3	6	Pass
short	4	12	Fail
short	5	2	Fail
short	5	10	Fail
long	1	2	Pass
short	1	0	Pass

- (a) From a decision tree perspective, calculate the information gain for all features and pick the best split. Tabulate the results. **(4 marks)**
- (b) From a random forest perspective, discuss how bagging and random splitting helps in reducing computational cost and avoid overfitting. Demonstrate the outcome using the dataset as in (a) and compare with the decision tree in (a). **(4 marks)**
- (c) Explain how ensemble methods can outperform decision tree and random forest. Suggest an example and visualize it. **(2 marks)**

END